

Q1. A rod OA in the figure rotates in the horizontal plane such that $\theta = t^3$ (in radian). At the same time, collar B is sliding outward along OA so that $r = 100t^2$ (mm). If in both cases t is in seconds. Determine the velocity and the acceleration of the collar when $t = 1$ second.

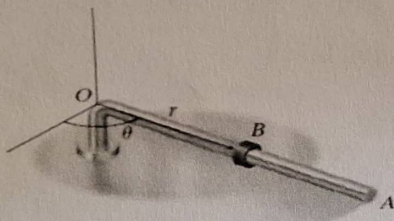


Fig 1.

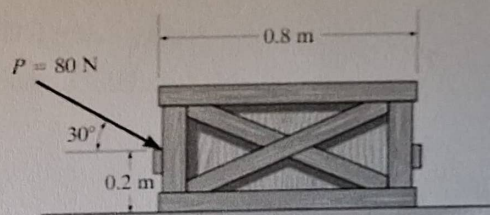


Fig 2.

Q2. The uniform crate shown in Fig. 2 has a mass of 20 kg. If a force $P = 80$ N is applied to the crate, determine if it remains in equilibrium. The coefficient of static friction is $\mu_s = 0.3$.

National Institute of Technology Goa
Engineering Mechanics (ME-100)

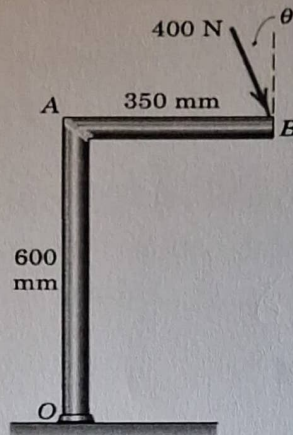
Minor 1, Date: Feb 24, 2026

Timing: 11:00-11:45 PM

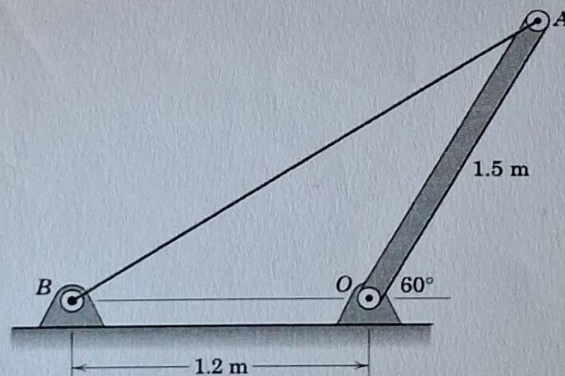
II Semester, Even 2025-26

Max mark: 20

1. A 400-N force is applied to the welded slender bar at an angle $\theta = 20^\circ$. Determine the equivalent force-couple system acting on the weld at (a) point A and (b) point O . For what value of θ would the results of parts (a) and (b) be identical? **6 marks**

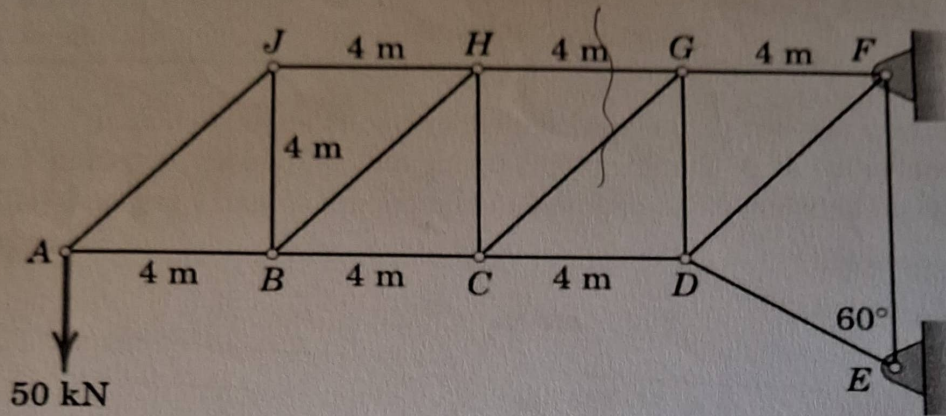


2. The uniform 18-kg bar OA is held in the position shown by the smooth pin at O and the cable AB . Determine the tension T in the cable and the magnitude and direction of the external pin reaction at O . **8 marks**



3. Determine the forces in members GH and CG for the truss loaded and supported as shown. Does the static indeterminacy of the supports affect your calculation?

6 marks



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National Institute of Technology Goa

Programme Name: **B.Tech, I year**

Minor II Examinations, April 2026

Course Name: **Engineering Mechanics**

Date: **23/04/2026**

Duration : **40 min**

Course Code: **ME100**

Time: **10:00 AM - 11:00 AM**

Max. Marks: **20**

Use Neat handwriting

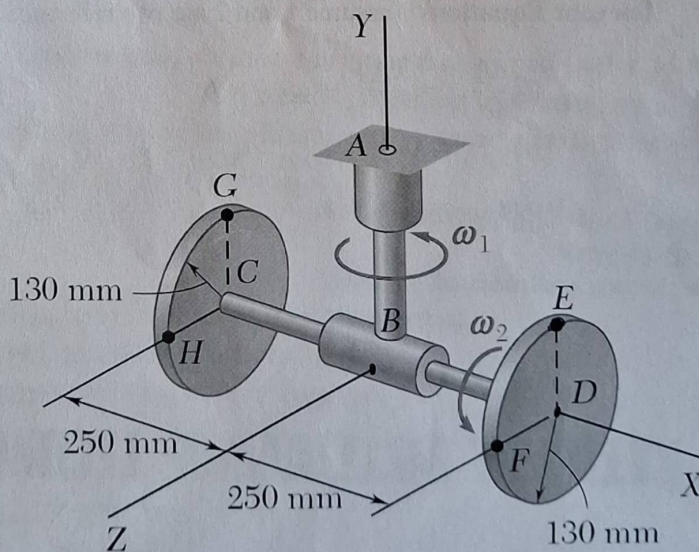
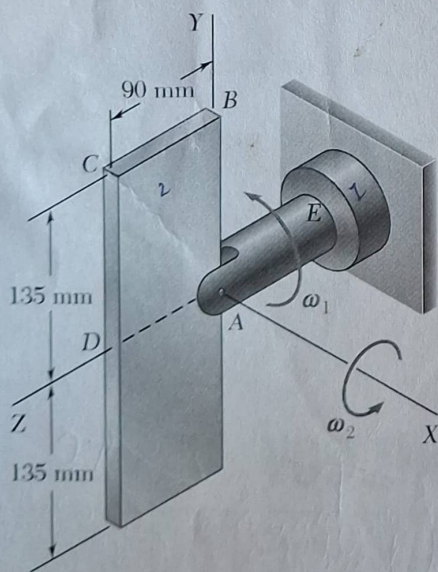
Problem 1: The rectangular plate (frame 2) shown rotates at the constant rate ω_2 rad/s with respect to arm AE (frame 1), which itself rotates at the constant rate ω_1 rad/s about the Z axis (see Figure 1). For the position shown, determine the velocity and acceleration with respect to ground of

- Point A
- Point B
- Point C
- Point D.

Problem 2: Two disks, each of 130 mm radius, are welded to the 500 mm rod CD . The rod-and-disks unit (see Figure 2) rotates at the constant rate $\omega_2 = 3$ rad/s with respect to arm AB (frame 1). Knowing that at the instant shown $\omega_1 = 4$ rad/s, determine the velocity and acceleration with respect to ground of

- Point E
- Point H

[8+6]



Problem 3: Attempt the following

a) Think of a point moving in a plane. Assume $(\underline{e}_t, \underline{e}_n)$ denote the unit vectors tangent and normal to the curve while $\underline{e}_b = \underline{e}_t \times \underline{e}_n$. Which of the following is true?

- i) All the unit vectors $(\underline{e}_t, \underline{e}_n, \underline{e}_b)$ are confined to the plane of motion
- ii) Only tangent vectors and normal vectors lie in the plane of motion
- iii) The vector $\underline{e}_b$ changes continuously along the path
- iv) none of these

[2]

b) Think of four rigid bodies denoted by $i = 1, 2, 3, 4$. Fill in the blanks

$$(i) \underline{\omega}_{4/1} = \underline{\omega}_{4/3} + \underline{\omega}_{3/2} + \dots$$

$$(ii) \underline{\alpha}_{4/1} = \dots$$

[1+2]

c) For a planar motion, $\frac{d\underline{e}_n}{ds}$ equals

$$(i) \underline{e}_b \times \underline{e}_t$$

$$(ii) \frac{\underline{e}_b \times \underline{e}_n}{\rho}$$

$$(iii) \underline{e}_b$$

$$(iv) \underline{0}$$

[1]

Relevant Equations: (assume 1 and 2 are two reference frames)

$$(i) \left. \frac{d\underline{A}}{dt} \right|_1 = \left. \frac{d\underline{A}}{dt} \right|_2 + \underline{\omega}_{2/1} \times \underline{A}$$

$$(ii) \underline{v}_{P/1} = \underline{v}_{P/2} + \underline{v}_{O'/1} + \underline{\omega}_{2/1} \times \underline{r}_{P/O'}$$

$$(iii) \underline{a}_{P/1} = \underline{a}_{P/2} + \underline{a}_{O'/1} + 2\underline{\omega}_{2/1} \times \underline{v}_{P/2} + \underline{\omega}_{2/1} \times (\underline{\omega}_{2/1} \times \underline{r}_{P/O'}) + \underline{\alpha}_{2/1} \times \underline{r}_{P/O'}$$



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: **B.Tech, I year**

Mid Semester Examinations, October 2025

Course Name: **Engineering Mechanics**

Date: **11/10/2025**

Duration : **90 min**

Course Code: **ME100**

Time: **2:00 PM - 3:30 PM**

Max. Marks: **50**

Answer all questions and use neat handwriting

Problem 1: Define principle of transmissibility of forces with an example. For the vectors $\vec{V}_1$ and $\vec{V}_2$ shown in the figure 1,

(a) determine $\vec{S} = \vec{V}_1 + \vec{V}_2$, $\vec{D} = \vec{V}_1 - \vec{V}_2$

(b) determine the angle between $\vec{D}$ with the positive x-axis .

Use vector mechanics to solve the problem.

[05]

Problem 2: In the design of the robot to insert the small cylindrical part into a close-fitting circular hole, the robot arm must exert a 90 N force P on the part parallel to the axis of the hole as shown in figure 2. Determine the components of the force which the part exerts on the robot along axes (a) parallel and perpendicular to the arm AB, and (b) parallel and perpendicular to the arm BC. [10]

Problem 3: At what angle must the 800 lb force be applied in order that the resultant **R** of the two forces have a magnitude of 2000 lb? For this condition, determine the angle between **R** and the vertical (see Figure 3) . [05]

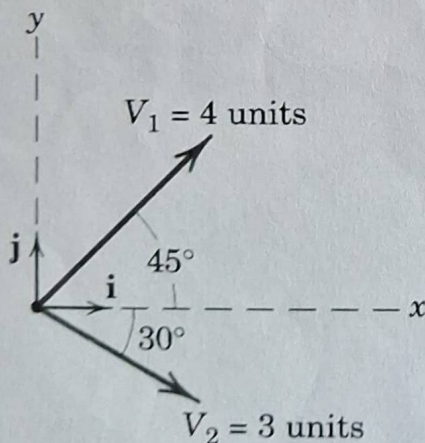


Figure 1

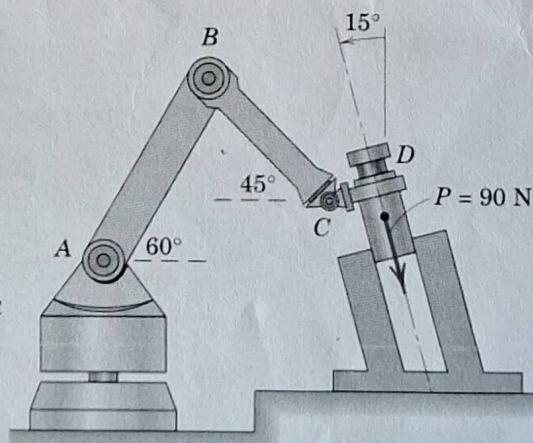


Figure 2

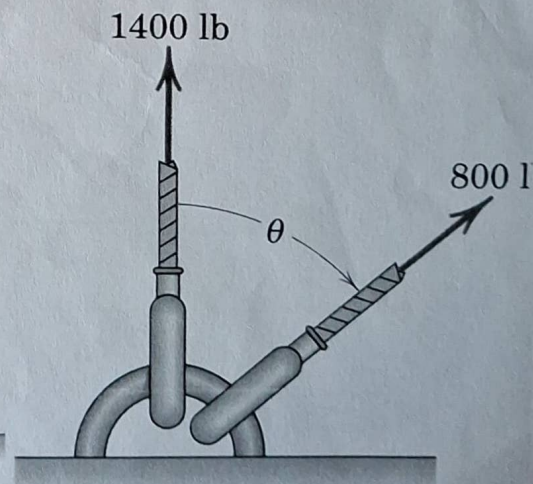


Figure 3

Problem 4: A commercial airliner (Figure 4) with four jet engines, each producing 90 kN of forward thrust, is in a steady, level cruise when engine number 3 suddenly fails. Determine and locate the resultant of the three remaining engine thrust vectors. Treat this as a two dimensional problem. [05]

Problem 5: The gear reducer (see Figure 5) shown is acted on by the two couples and its weight of 200 N. To design adequate mounts, determine the vertical force supported by each of the reducer mountings at A and B. [05]

Problem 6: For the loaded truss shown in figure 6, set up algebraic equations to find forces in all the members of the truss. All members have equal lengths ($AB=1\text{m}$). DO NOT SOLVE THE EQUATIONS. [10]

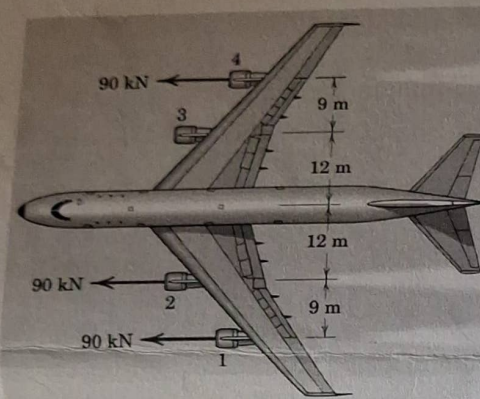


Figure 4

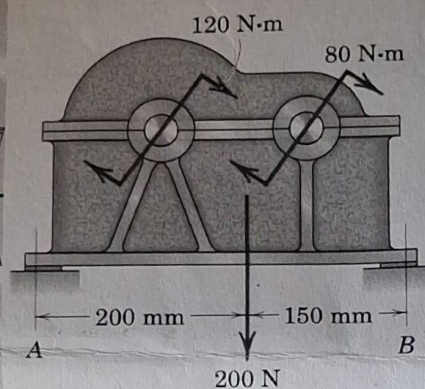


Figure 5

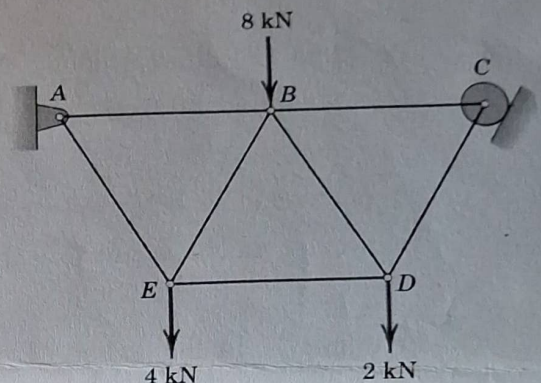


Figure 6

Problem 7: Two forces act on the rod shown in Figure 7. Determine the resultant moment they create about the flange at O (reference point). Use vector mechanics to solve the problem. [05]

Problem 8: Replace the force and couple system shown in Figure 8 by an equivalent resultant force and couple moment acting at point O. Use vector mechanics to solve the problem. [05]

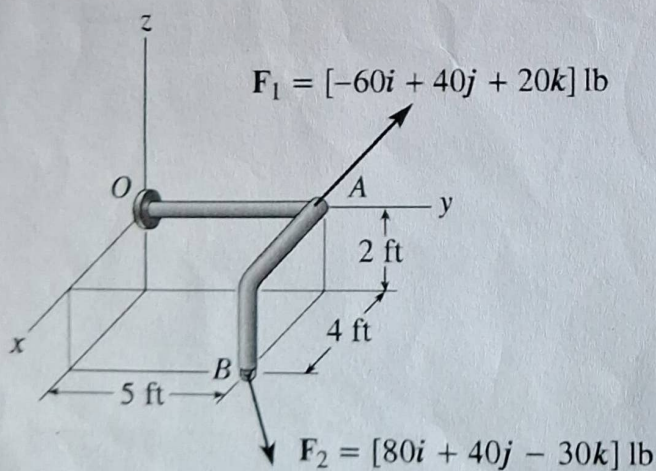


Figure 7

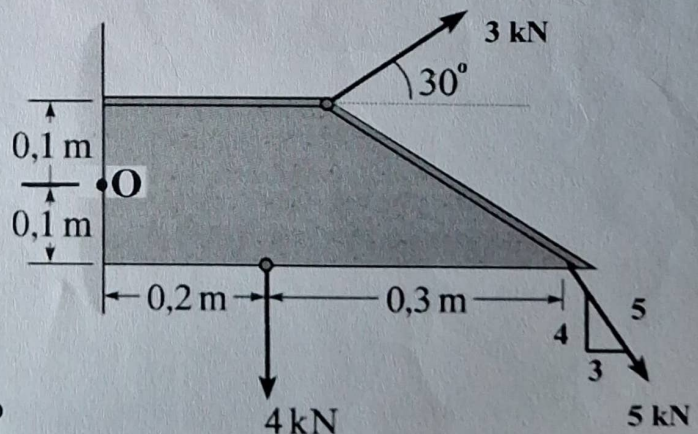


Figure 8

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NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: **B.Tech, I year**

End Semester Examinations, May 2026

Course Name: **Engineering Mechanics**Date: **07/05/2026**Duration : **180 min**Course Code: **ME100**Time: **10:00 AM - 1:00 PM**Max. Marks: **100**

Use neat handwriting

1. State the necessary conditions for the equilibrium of a rigid body in a plane. Clearly explain how these conditions differ for concurrent and non-concurrent force systems. Two cylinders of diameters 100 mm and 50 mm, weighing 200 N and 50 N, respectively are placed in a trough (see Figure 1). Neglecting friction, find the reactions at contact surfaces 1, 2, 3 and 4. [10]
2. Explain the physical significance of the moment of inertia of an area. How does the orientation of the area affect its moment of inertia about different axes? A thin uniform semicircular plate (Figure 2) of radius a is placed in the first quadrant such that its diameter is inclined at 45° to the X -axis. Determine the following [10]
 - (a) Coordinates of the centroid $(\bar{x}, \bar{y})$ of the shaded area with respect to the X - and Y -axes.
 - (b) Moment of inertia of the shaded area about the X -axis and Y -axis.
3. Two identical blocks A and B (Figure 3) are connected by a rod and they rest against vertical and horizontal planes respectively as shown in Fig.4. If sliding impends when $\theta = 45^\circ$, determine the coefficient of friction, assuming it to be the same for both floor and wall. [10]

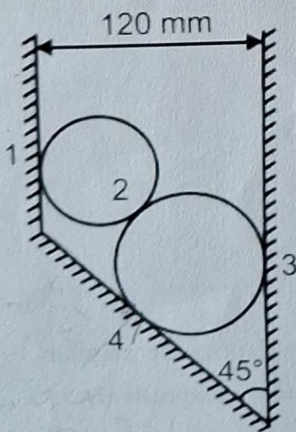


Figure 1

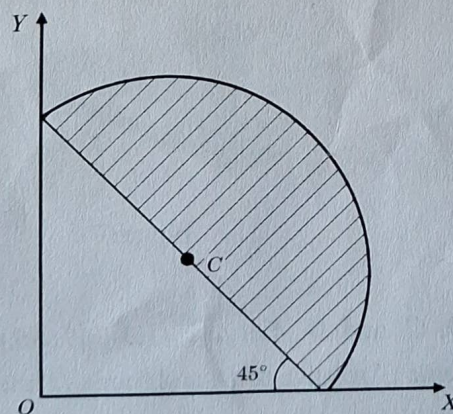


Figure 2

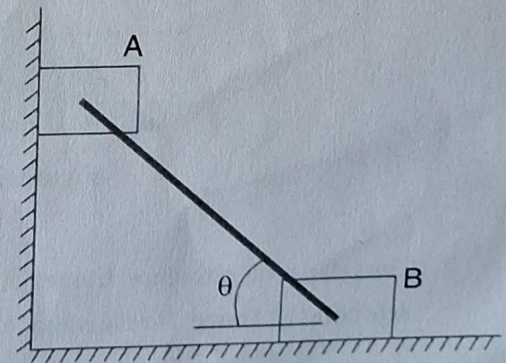


Figure 3

4. Determine the force in each member of the loaded truss (Figure 4).

[06]

5. A bracket is subjected to three forces and a couple as shown in Figure 5. Determine magnitude, direction and the line of action of the resultant. Also discuss under what condition the system reduces to a pure couple only.

[04]

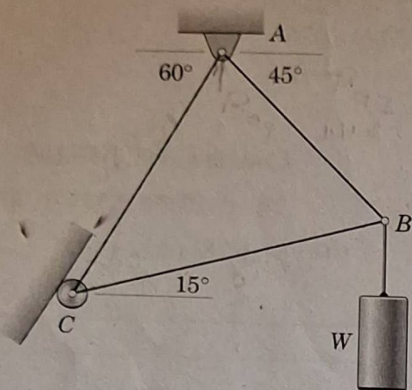


Figure 4

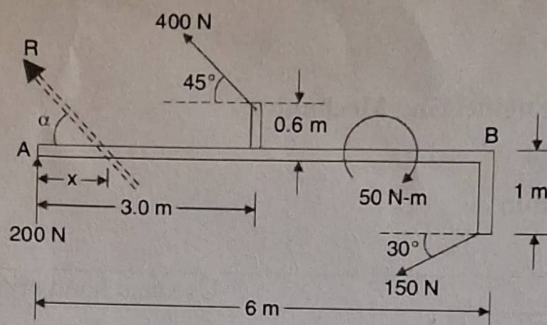


Figure 5

6. As rod OA rotates, pin P moves along the parabola BCD (Figure 6). Knowing that the equation of this parabola is $r = \frac{2b}{[1 + \cos(\theta)]}$ and that $\theta = kt$, determine the velocity and acceleration of P with respect to ground when (i) $\theta = 0^\circ$, (ii) $\theta = 45^\circ$, (iii) $\theta = 90^\circ$.

[10]

7. The circular disk (Figure 7) is spinning about its own axis (y-axis) at the constant rate $p = 10\pi$ rad/s. Simultaneously, the frame is rotating about the Z-axis at the constant rate $\Omega = 4$ rad/s. Calculate the **angular** acceleration of the disk and the acceleration of point A at the top of the disk with respect to ground.

[10]

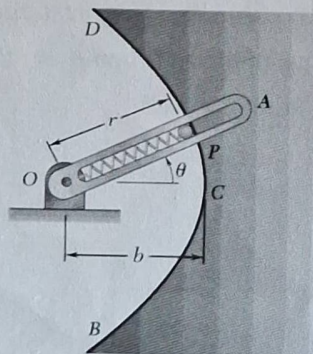


Figure 6

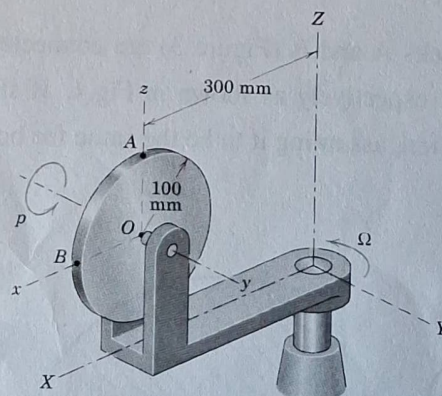


Figure 7

8. Consider four reference frames A , B , C , and D . Let $\vec{\omega}_{B/A}$, $\vec{\omega}_{C/B}$, and $\vec{\omega}_{D/C}$ denote the angular velocities of frame B relative to A , frame C relative to B , and frame D relative to C , respectively.

(a) Write the expression for the angular velocity of frame D relative to frame A .

(b) Also derive relation for the angular acceleration $\vec{\alpha}_{D/A}$ in terms of $\vec{\alpha}_{B/A}$, $\vec{\alpha}_{C/B}$, and $\vec{\alpha}_{D/C}$. [08]

9. In the position shown (see Figure 8), bar DE has a constant angular velocity of 50 rad/s clockwise. Knowing that $h = 500 \text{ mm}$, determine (a) the angular velocity of bar FBD , (b) the velocity of Point F with respect to ground. [08]
10. Derive an expression for jerk in cylindrical coordinate system. [08]
11. Determine the constant force P that will give the system of bodies (see Figure 9) a velocity of 3 m/sec after moving 4.5 m from rest. Coefficient of friction between the blocks and the plane is 0.3 . Pulleys are smooth. Use work energy method. [08]
12. The three small spheres (Figure 10) are welded to a light rigid frame which is rotating in a horizontal plane about a vertical axis through O with an angular velocity $\dot{\theta} = 20 \text{ rad/s}$. If a couple $M_O = 30 \text{ N}\cdot\text{m}$ is applied to the frame for 5 seconds, compute the new angular velocity $\dot{\theta}$. [08]

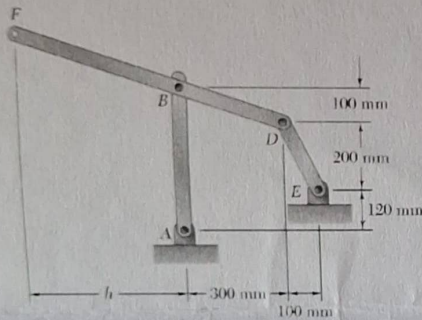


Figure 8

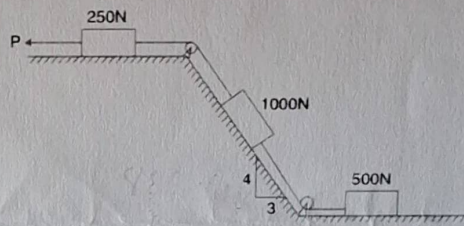


Figure 9

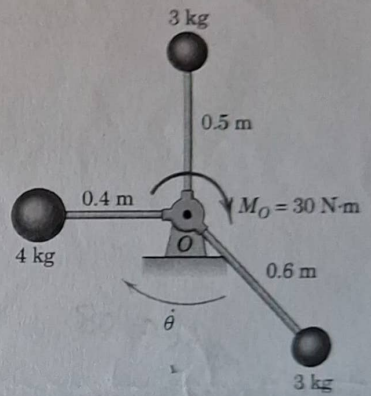


Figure 10

END

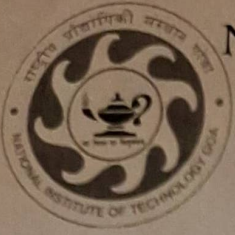
Relevant Equations: (assume 1 and 2 are two reference frames)

$$(i) \left. \frac{d\vec{A}}{dt} \right|_1 = \left. \frac{d\vec{A}}{dt} \right|_2 + \vec{\omega}_{2/1} \times \vec{A}$$

$$(ii) \vec{v}_{P/1} = \vec{v}_{P/2} + \vec{v}_{O'/1} + \vec{\omega}_{2/1} \times \vec{r}_{P/O'}$$

$$(iii) \vec{a}_{P/1} = \vec{a}_{P/2} + \vec{a}_{O'/1} + 2\vec{\omega}_{2/1} \times \vec{v}_{P/2} + \vec{\omega}_{2/1} \times (\vec{\omega}_{2/1} \times \vec{r}_{P/O'}) + \vec{\alpha}_{2/1} \times \vec{r}_{P/O'}$$

$$(iv) \vec{\alpha}_{3/1} = \vec{\alpha}_{3/2} + \vec{\alpha}_{2/1} + \vec{\omega}_{2/1} \times \vec{\omega}_{3/2}$$



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: **B.Tech, I year**

End Semester Examinations, December 2025

Course Name: **Engineering Mechanics**

Date: **12/12/2025**

Duration : **180 min**

Course Code: **ME100**

Time: **10:00 AM - 01:00 PM**

Max. Marks: **100**

Answer Part A, Part B, and Part C **separately**, clearly stating any **assumptions** made for missing data.

PART A

- Derive, from first principles, the expression for the centroid of a semicircle. Also, compute the moment of inertia of the shaded region (dimensions in mm) shown in Figure 1 about axis AB, applying the parallel-axis theorem where required. [04+08]
 - Two identical blocks A and B are connected by a rod and they rest against vertical and horizontal planes respectively as shown in Figure 2. If sliding impends when $\theta = 45^\circ$, determine the coefficient of friction, assuming it to be the same for both floor and wall. [08]
 - Explain Lami's Theorem and Varignon's Theorem, clearly stating what each theorem says and where it is applied. [04]
- A bracket is subjected to three forces and a couple as shown in Figure 3. Determine magnitude, direction and the line of action of the resultant. [08]

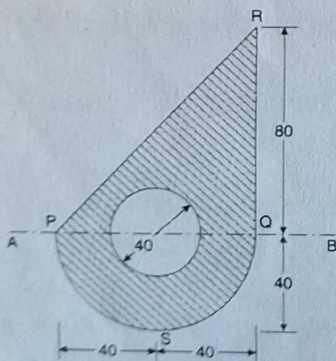


Figure 1

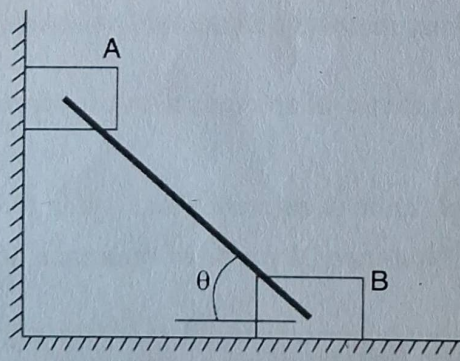


Figure 2

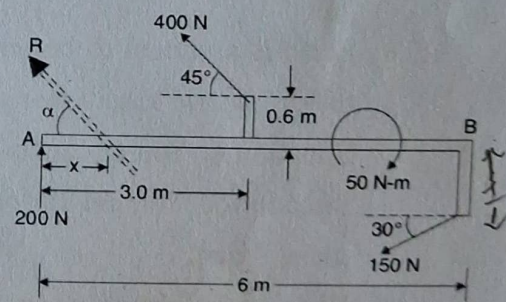


Figure 3

- The resultant of two forces, one of which is double the other is 260 N. If the direction of larger force is reversed and the other remains unaltered, the resultant reduces to 180 N. Determine the magnitude of the forces. [04]

PART B

5. The circular disk (Figure 4) is spinning about its own axis (y-axis) at the constant rate $p = 10\pi$ rad/s. Simultaneously, the frame is rotating about the Z-axis at the constant rate $\Omega = 4$ rad/s. Calculate the angular acceleration of the disk and the acceleration of point A at the top of the disk. [12]
6. The pin A of the hinged link AC is confined to move in the rotating slot of link OD (see Figure 5). The angular velocity of OD is $\omega = 2$ rad/s clockwise and is constant for the interval of motion concerned. For the position where $\theta = 45^\circ$ with AC horizontal, determine the velocity of pin A relative to ground. [08]

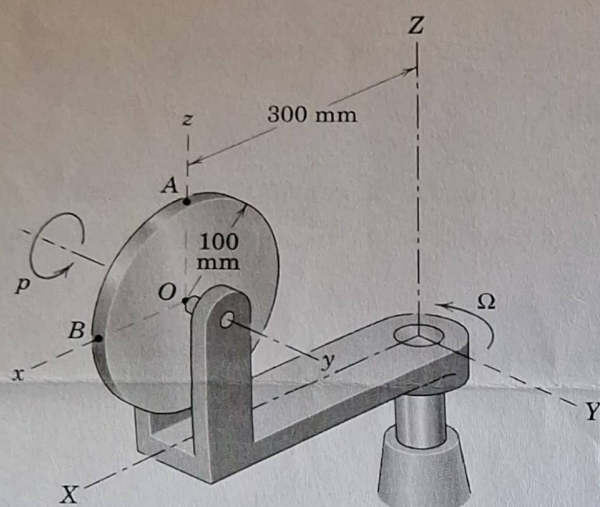


Figure 4

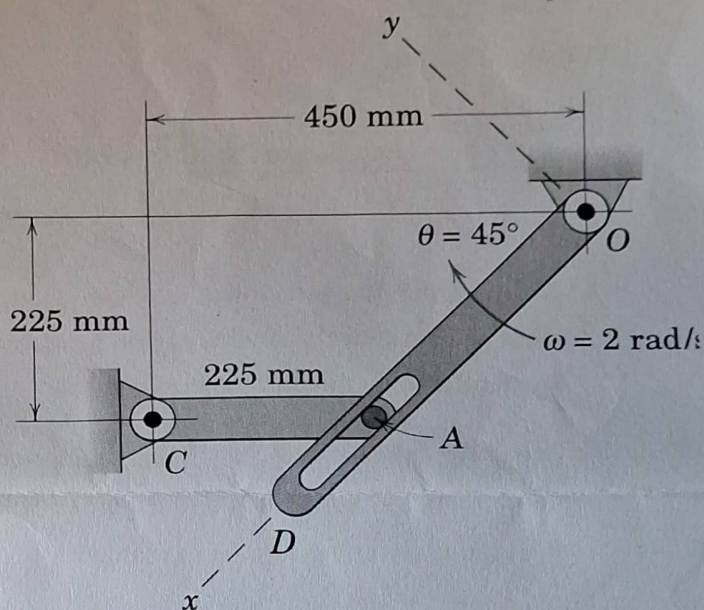


Figure 5

7. Consider three frames of reference: A, B, and C. Let $\vec{\omega}_{B/A}$, $\vec{\omega}_{C/A}$, and $\vec{\omega}_{C/B}$ be the angular velocities of frame B relative to frame A, frame C relative to frame A, and frame C relative to frame B, respectively. Derive an equation that shows the relationship between their angular accelerations. [08]
8. Earth is assumed to be an Inertial frame of reference for all practical purposes. Now, imagine the following two situations:
 Situation 1: You are on an Inertial frame of reference and trying to observe an airplane flying.
 Situation 2: You are sitting on a Non-inertial frame of reference and trying to observe the same airplane flying.
 With this information, answer the following
 (i) Derive an expression that shows the relationship between the acceleration of the airplane with respect to the Inertial and Non-inertial frame of reference?
 (ii) Write an expression for Newton's Second law of motion for Situation 1 and Situation 2 respectively. [04+02]

PART C

9. Derive an expression for jerk in cylindrical coordinate system. [06]
- An electron moving with initial velocity $(v_0 \vec{e}_1)$ is subjected to an acceleration $\frac{e E_0}{m} \sin(\omega t)$ at right angles to $(v_0 \vec{e}_1)$. Find the position vector at time t and trajectory of the electron. [08]
10. As rod OA rotates (Figure 6), pin P moves along the parabola BCD . Knowing that the equation of this parabola is $r = \frac{2b}{[1 + \cos(\theta)]}$ and that $\theta = kt$, determine the velocity and acceleration of P with respect to ground when (i) $\theta = 0^\circ$, (ii) $\theta = 45^\circ$ (iii) $\theta = 90^\circ$. [08]
11. Determine the constant force P that will give the system of bodies (see Figure 7) a velocity of 3 m/sec after moving 4.5 m from rest. Coefficient of friction between the blocks and the plane is 0.3. Pulleys are smooth. Use work energy method. [08]

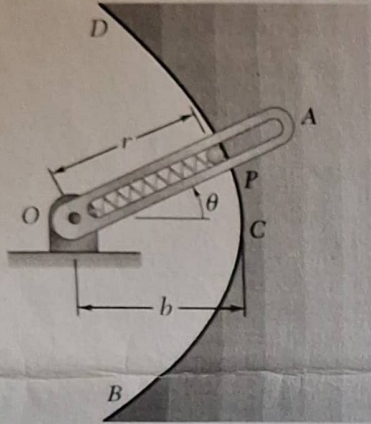


Figure 6

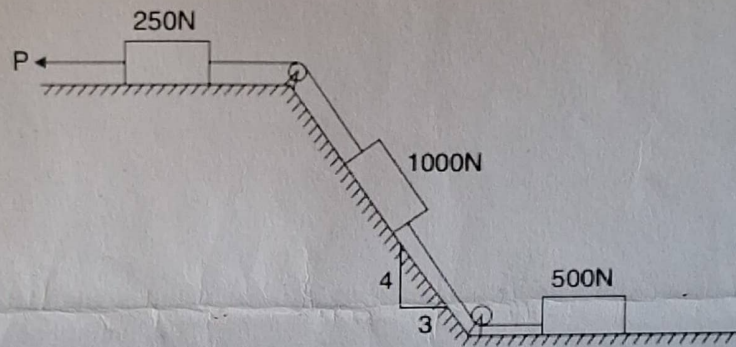


Figure 7

Relevant Equations: (assume 1 and 2 are two reference frames)

$$(i) \left. \frac{d\vec{A}}{dt} \right|_1 = \left. \frac{d\vec{A}}{dt} \right|_2 + \vec{\omega}_{2/1} \times \vec{A}$$

$$(ii) \vec{v}_{P/1} = \vec{v}_{P/2} + \vec{v}_{O'/1} + \vec{\omega}_{2/1} \times \vec{r}_{P/O'}$$

$$(iii) \vec{a}_{P/1} = \vec{a}_{P/2} + \vec{a}_{O'/1} + 2\vec{\omega}_{2/1} \times \vec{v}_{P/2} + \vec{\omega}_{2/1} \times (\vec{\omega}_{2/1} \times \vec{r}_{P/O'}) + \vec{\alpha}_{2/1} \times \vec{r}_{P/O'}$$

END